Trailing Labor Yield: A White Paper

Version: v0.9 public draft
Status: mechanism design proposal
Repository references: `sim/`, `dao/contracts/`

Abstract

Trailing Labor Yield (TLY) is a historical-compensation-indexed trailing labor claim: a compensation architecture in which past compensated contribution determines a finite, tapering stream of post-exit cash flows. Its components resemble familiar instruments: deferred compensation, phantom participation, contingent benefits, and runoff liabilities. Its novelty is architectural. It links lifetime eligible compensation history, finite runoff, funding discipline, and production governance into one modelable structure. The stablecoin- and EVM-based design presented here is a reference implementation of that broader structure, chosen because reserve assets, programmable settlement, and per-wallet claim accounting fit the mechanism naturally. This first paper is concept-first, with formal mechanism sections where the mechanism needs them and practical deployment layers where treasury, governance, and legal choices change the worker-side promise. TLY is not appropriate for every organization; it needs reserve planning, compensation discipline, and real legal review.

Executive Summary

- TLY is a historical-compensation-indexed trailing labor claim: compensated contribution history determines a finite stream of post-exit cash flows.
- Pure TLY defines the mathematical mechanism: active pay, active bonus accrual, a lifetime eligible-compensation base, and a tapering claim over a fixed term.
- The Stress Layer defines whether the claim is prudent: affordability, reserve coverage, payout priority, pause/queue/partial-payment rules, and missed-epoch treatment.
- The Governance Wrapper defines production safety: eligible-compensation rules, ledger integrity, exceptional compensation review, contributor-admin controls, and parameter control.
- The stablecoin/EVM design is a reference implementation for programmable settlement, not a complete legal, tax, accounting, employment, or production deployment package.

Who This Is For

This paper is written for founders, DAO operators, contributor pods, protocol treasurers, compensation committees, and workers negotiating long-term upside. It assumes basic familiarity with stablecoins, treasury management, and smart contract-based payroll. It does not assume that the

reader wants to start in the Solidity or the simulator. The mechanism should make sense before any of that.

Thesis And Scope

The claim is architectural. TLY combines familiar economic components into a specific compensation structure: compensated labor history determines a finite, tapering post-exit cash-flow claim. That structure can be analyzed apart from any one implementation, but the stablecoin/EVM version is a strong first deployment surface because programmable settlement, reserve assets, and per-wallet claim accounting fit the mechanism naturally.

TLY is not automatically better than equity, RSUs, phantom equity, token vesting, or a normal deferred-bonus pool. It is most interesting where an organization wants modelable economic continuation without turning governance tokens into the compensation sink.

This draft is concept-first. It formalizes the mechanism where precision matters, but it does not pretend to settle every legal wrapper, tax treatment, accounting treatment, or production treasury policy. Those choices are not decorations. They change what the worker has actually been promised.

The paper uses three layers:

1. Pure TLY: the mathematical mechanism only.
2. Stress Layer: affordability, reserve coverage, payout-priority rules, pause semantics, and missed-epoch treatment.
3. Governance Wrapper: eligible-compensation rules, compensation-ledger integrity, exceptional compensation review, contributor-admin controls, and parameter control.

Plain-English Mechanism

TLY pays contributors normal cash compensation while they are active. It also grants a small bonus tied to a historical compensation base. When a contributor leaves, the same cumulative compensation base determines the initial trailing cash-flow claim for a fixed number of periods. In the reference implementation, that claim is paid in a stablecoin or treasury reserve asset. The payout tapers over time, so former contributors keep some upside without leaving an ever-growing bill behind them.

The mechanism has two states:

1. Active state: the contributor receives base pay and earns a compounding active bonus.
2. Legacy state: the contributor no longer earns base pay, but can claim a trailing payout based on lifetime eligible compensation at the company.

TLY pays in a treasury reserve asset, such as a stablecoin, rather than a governance token. That choice does a lot of work. Too many token-compensation schemes quietly ask the same asset to govern,

signal, collateralize, reward, and absorb constant sell pressure. In practice, that bundle is often brittle.

Mechanism At A Glance

If someone wants the short version, it is just this:

1. While active, the contributor receives base pay and earns an active bonus.
2. On exit, the protocol uses the contributor's cumulative eligible compensation base to define the initial trailing amount.
3. In legacy state, that amount is paid out through a tapering claim schedule over a fixed duration.

A Worked Example

Consider one contributor paid annually.

Input	Value
Base pay	$100,000 per year
Historical compensation base entering year	$250,000
Active accrual share	1%
Annual taper after exit	5%
Trailing duration	25 years

The active bonus for the year is:

$$\text{Active Bonus} = (\$250,000 + \$100,000) \times 0.01 = \$3,500\,.$$

If the contributor exits after that year, the reference convention uses the accrual share applied to the contributor's eligible lifetime compensation base as the initial trailing amount. Under the non-self-referential timing convention used here, the exit base for this year is prior historical compensation plus current base pay, so the amount is the same as the cached active-bonus amount:

$$R = \$3,500\,.$$

This paper uses the same timing convention as the contracts: the first claimable trailing payout occurs one full epoch after departure and equals $R\rho$, not R.

With a 5 percent annual taper, the first three trailing payouts are:

Year after exit	Payout
1	$3,325.00

Year after exit	Payout
2	$3,158.75
3	$3,000.81

The total undiscounted payout over 25 years is approximately:

$$\sum_{s=1}^{25} 3500 \times 0.95^{s} \approx \$48,054\,.$$

This small example is useful, but it is also where a clean formula can become dangerous. Bounded means the obligation runs off structurally rather than compounding forever; it does not mean every parameter set is prudent or fully prefunded. That distinction matters. People hear “bounded” and often smuggle in “affordable.” Those are not the same claim. The organization-level burden still depends on payroll growth, turnover, active accrual rate, trailing duration, and the taper.

The Compensation Problem

Traditional startup equity solves cash preservation for the company but often fails workers. The worker receives paper exposure to a private capital structure with uncertain liquidity, exercise costs, preferred-share overhang, and limited control. The claim is called ownership, but for many workers it behaves like a long-dated option with unclear realizable value.

DAO token compensation improves liquidity but introduces reflexivity. If contributors receive governance tokens and sell them for living expenses, the protocol pays labor by pressuring the same asset that coordinates governance. If the token price falls, more tokens are required to pay the same real wage. That can create a procyclical compensation loop.

Pensions and perpetuities fail in the opposite direction. They reduce worker uncertainty by creating employer uncertainty. Fixed, permanent obligations can accumulate into a large inactive claimant base. This is the Iceberg Effect: the visible current payroll is only the surface; the hidden mass is the stock of future obligations.

TLY sits somewhere between these structures. It gives workers a continuing economic right after exit, but not an untapered one and not an ownership claim.

Layer 1: Pure TLY

Pure TLY is the compensation rule. It says how active compensation updates the historical compensation base, how the initial trailing amount is defined from that base, and how that amount runs off after departure. It does not answer whether the organization can afford the claim, whether

paused claims queue, or what legal form the promise takes. Those questions belong to the Stress Layer and Governance Wrapper.

What TLY Is

TLY is a compensation rule that turns compensated labor history into a trailing payout right.

It is not equity because it does not transfer governance or ownership. It is also not ordinary token compensation, because the mechanism does not depend on workers selling the governance asset into the market. And it is not a standard pension. The payout is tied to the contributor's compensation history, begins from the accrued lifetime compensation base, and then tapers over a fixed horizon.

TLY gives contributors a continuing cash-flow claim if the organization survives and keeps funding it. That is real upside, but it is not the same thing as owning an open-ended equity position.

In the reference design, the active bonus is:

$$\text{Active Bonus} = (\text{Historical Compensation Base} + \text{Base Pay}) \times \text{Accrual Share}\,.$$

In the reference design, the initial trailing amount is the accrual share applied to the contributor's eligible lifetime compensation base at exit. Under the timing convention used here, that base is prior historical compensation plus current-period eligible base pay; the current active bonus is then added to the historical compensation base for future active periods if the contributor continues. The contracts cache the same quantity as `lastActiveBonus`, but the economic object is not a discretionary terminal bonus; it is the accumulated compensation ledger multiplied by the accrual share. The trailing claim then pays:

$$\text{Trailing Payout}_s = \text{Initial Trailing Amount} \times \rho^s,$$

where s is the number of epochs after departure and ρ is the per-epoch taper factor. For annual epochs and a 5 percent taper, $\rho = 0.95$.

Pure TLY does not decide whether that cash-flow claim is legally enforceable deferred compensation, a reserve-backed benefit, or a contingent protocol benefit. The same formula can sit under different wrappers, and those wrappers matter.

Historical Compensation Base And Compounding

The historical compensation base is the core state variable on the active side. In the contracts it is named `historicalPayPool`, but it is an accounting/state variable, not necessarily a segregated pool of funded assets. This is also the place where many readers quite reasonably stop and ask, "Wait, what is actually compounding here?" In the reference design, the base is not just cumulative base salary. It is cumulative settled active compensation that remains attached to the active workforce.

At the contributor level, while a contributor remains active:

$$H_{i,t+1} = H_{i,t} + B_{i,t} + A_{i,t}\,.$$

At the aggregate level, after turnover is applied:

$$H_{t+1} = (H_t + B_t + A_t)(1 - q)\,.$$

This means prior active bonuses enlarge the future bonus base. That compounding is intentional. In TLY, the historical compensation base behaves more like a running labor-credit account than a simple wage ledger. The mechanism is trying to reward accumulated compensated contribution, not only elapsed tenure.

A narrower design could compound only on prior base pay and ignore past active bonuses. That would be a different and less aggressive mechanism. It is not the reference system simulated in `sim/` or implemented in the Solidity contracts.

There are at least three plausible variants:

Variant	Compensation base	Tradeoff
Full recursive design	Base pay plus prior settled active bonuses	Strongest continuation claim, but most sensitive to stationarity assumptions
Base-pay-only design	Base pay only	Easier to reason about, but weaker reward for accumulated compensated contribution
Hybrid design	Base pay plus capped or partially weighted settled bonuses	More governance discretion, but potentially easier to fit to treasury policy

This paper uses the full recursive design as the reference mechanism. That choice should be visible rather than smuggled into the equations. Prior settled active bonuses enlarge the future compensation base; if that feels like a small detail, it is not.

Bounded Runoff And Stationarity

This is the place where people most easily overclaim. Tapering the legacy side is necessary, but it does not save the model if the active-side historical compensation base is allowed to outrun payroll indefinitely. The old claims have to lose weight, and the active bonus base has to remain tied to payroll rather than becoming its own runaway ledger.

For a trailing claim with initial amount R, taper factor $\rho < 1$, and duration K, total undiscounted payout is:

$$\sum_{s=1}^{K} R\rho^s = R\rho\frac{1-\rho^K}{1-\rho}\,.$$

At a 5 percent annual taper and 25-year duration:

$$\rho = 0.95,$$

and:

$$R \times 0.95 \times \frac{1 - 0.95^{25}}{0.05} \approx 13.73R\,.$$

This number is the lifetime multiple of the initial trailing amount, not of base salary. In the minimal reference design, that initial amount is controlled by the active accrual share, typically modeled at 1 percent.

Tapering alone is not the whole stationarity story. Let G denote gross payroll growth from one epoch to the next, let q denote turnover, and define the aggregate historical-pool ratio:

$$h = \frac{H_t}{B_t}\,.$$

From the aggregate update equations:

$$H_{t+1} = (H_t + B_t + A_t)(1 - q)$$

and

$$A_t = \alpha(H_t + B_t),$$

the key stability parameter is:

$$c = \frac{(1 - q)(1 + \alpha)}{G}\,.$$

If $c < 1$, then the active-side state converges to a steady ratio:

$$h = \frac{c}{1 - c}$$

and the active bonus ratio converges to:

$$m = \frac{A_t}{B_t} = \frac{\alpha}{1 - c}\,.$$

If $c \geq 1$, the historical compensation base grows at least as fast as payroll and the active bonus ratio does not settle. So yes, tapering matters. But tapering by itself does not rescue a badly chosen active-side design.

At the organization level, four forces interact:

1. Low active accrual limits new trailing claims.
2. Payroll growth makes old cohorts smaller relative to current payroll.
3. The taper reduces the weight of every legacy cohort over time.
4. Turnover limits how much historical pay remains attached to active workers.

Under the simulator's baseline growth intuition of roughly 4 percent gross payroll growth and the EVM convention that the first legacy claim receives $R\rho$, the approximate steady burden is shown below.

The active bonus ratio column is not a free input. It comes out of the stationarity condition above. With $\alpha = 1\%$ and $G \approx 1.04$, the model yields $m \approx 9.35\%$ at 8 percent turnover, $m \approx 7.92\%$ at 10 percent turnover, and $m \approx 3.12\%$ at 30 percent turnover. That may look backward at first glance, but it makes sense once you sit with it: higher turnover means fewer contributors remain active long enough to build large historical compensation bases.

Turnover	Active bonus ratio	Approx. trailing burden
8%	9.35%	7.9% of payroll
10%	7.92%	8.3% of payroll
30%	3.12%	9.8% of payroll

These values are illustrative rather than universal. Boundedness is a mechanical property, not a solvency guarantee. The model stabilizes only when the active bonus base remains bounded relative to payroll and the trailing side actually runs off over time. No artificial cap on an individual contributor's historical compensation base is needed for that result, although the governance problem does not disappear. An administrator who can change compensation one epoch before departure can materially affect the claim.

Layer 2: Stress Layer

Pure TLY can be bounded and still be imprudent. The Stress Layer is where the organization decides whether the formula is affordable, what priority legacy claims have, and what happens when the treasury is under pressure. This is not secondary paperwork. It is the part that keeps a mechanical runoff property from being mistaken for a solvency guarantee.

Treasury-Health Metrics

At minimum, a DAO considering TLY should track:

- trailing burden / active payroll;
- trailing burden / gross margin or operating surplus;
- reserve coverage, measured as stable reserves divided by projected trailing claims over a fixed look-ahead window;
- forward claim coverage, measured as stable reserves plus committed funding divided by active pay plus projected legacy claims over the same window;
- active bonus ratio, measured as active bonuses divided by payroll;

- the stress threshold at which legacy claims are paused, partially paid, or queued.

These metrics are not ornamental dashboard widgets. They are the operating boundary of the promise. If the dashboard says the model is drifting out of range and governance keeps issuing claims anyway, the problem is no longer the formula.

Stress-State Semantics

Before deployment, the organization should publish what happens in stress. There are several different promises hiding under the phrase “pause legacy claims”:

Stress behavior	What it means	Worker-side promise
Pause	Legacy claims stop while active pay remains claimable	Weakest; claim is live only when treasury policy permits
Partial payment	Legacy claims receive a disclosed percentage or pro rata amount	More predictable, but requires allocation rules
Queue / catch-up	Missed claims remain recorded and can be paid later	Stronger worker promise; creates a larger treasury overhang
No catch-up	Missed epochs do not accrue beyond the claim rules	Cleaner solvency rule; harsher for alumni

There is no mechanism design trick that makes future cash obligations disappear. They are funded, subordinated, queued, partially paid, or left to unfunded assumptions. The paper can model the first four. It should not pretend the fifth is a policy.

Sensitivity And Stress Cases

The baseline table above is only a starting point. A publication-ready model should expose sensitivity to:

- accrual share;
- turnover;
- payroll growth;
- taper;
- trailing duration;
- revenue or margin assumptions;
- reserve level and reserve drawdown speed.

At least one stress case should be shown before adoption: a revenue shock, payroll contraction, turnover spike, reserve decline, or a combined bear-market case where several of those happen

together. That is where a mechanism that looked reasonable under smooth growth either survives contact with reality or starts asking governance to improvise.

Design Conditions

TLY is most defensible when the following conditions hold:

- low active accrual share;
- taper factor below one;
- finite trailing duration;
- active-side stationarity, meaning $c < 1$;
- credible treasury funding policy;
- published stress-state semantics;
- governance controls around eligible-compensation definitions and ledger integrity.

Layer 3: Governance Wrapper

The Governance Wrapper is the set of rules that keeps the clean mechanism from being gamed by people who understand exactly where the formula is sensitive. The pure mechanism is elegant, but production safety requires wrapper rules. A production TLY design should not let the eligible-compensation ledger become a soft governance variable. The initial trailing amount should be derived from lifetime eligible compensation multiplied by the accrual share, not from a salary number or discretionary bonus chosen near departure.

Production Compensation-Ledger Default

The production default should be:

$$R_i = \alpha E_{i,\tau_i},$$

where E_{i,τ_i} is the contributor's eligible lifetime compensation base at exit. That base may include base pay, settled active bonuses, or other earned compensation depending on the deployment design, but the definition has to be fixed before the claim is earned. In the reference recursive design, prior settled active bonuses enlarge the base; a base-pay-only variant would be cleaner but less generous.

This is why ordinary terminal-period manipulation should not be the main risk in the model. A contributor should not be able to improve the trailing claim by nudging one terminal salary number. The real governance risk is upstream: whether administrators can edit the compensation ledger, reclassify exceptional payments as eligible compensation, or change parameters after contributors have already relied on the rule.

Recommended wrapper rules include:

- a disclosed definition of eligible compensation;

- append-only or auditable compensation-ledger updates;
- separation between ordinary compensation updates and exceptional-comp review;
- governance review for exceptional compensation changes;
- published parameter-change procedures;
- periodic reporting of active payroll, active bonuses, projected legacy burden, and reserve coverage.

Reference Implementation Versus Production Design

The current Solidity reference stores `lastActiveBonus` for simplicity and gas economy. In this design, `lastActiveBonus` is the cached result of applying the accrual share to the contributor's historical compensation base for the last settled active epoch. It should not be read as a discretionary terminal-period bonus. Production deployments should still harden compensation-ledger updates, exceptional-comp review, role controls, and parameter-change procedures.

Operational Design And Governance Rules

The reference EVM architecture has two contracts:

Contract	Role
`ContributorRegistry.sol`	stores contributor status, base pay, historical compensation base (`historicalPayPool`), cached active bonus, departure epoch, and claim checkpoints
`TreasuryDistributor.sol`	holds the payment token, computes active compensation, computes trailing claims, and transfers funds

The system uses per-contributor accounting. The registry stores each contributor's `historicalPayPool`, `lastActiveBonus`, `initialLegacyAmount`, `departureEpoch`, and `lastLegacyClaimEpoch`.

The treasury uses pull payments:

- active contributors call `claimActiveComp()`;
- legacy contributors call `claimTrailingYield()`.

The treasury never iterates through all alumni. That is essential for gas safety.

The governance design separates roles:

- `POD_ADMIN_ROLE`: operational sub-DAO or multisig that can add contributors, remove contributors, and update base pay.
- `DAO_ADMIN_ROLE`: treasury-level role that can pause and unpause legacy claims during stress.

If legacy claims are paused, active compensation remains claimable. This keeps current labor funded during bear markets or treasury stress.

Worker-Side Promise Variants

The same TLY formula can support different promises. Workers should not have to infer which one they are being offered.

Variant	What the worker has	Main deployment burden
Contingent protocol benefit	A claim defined by protocol rules and treasury state	Clear disclosure of pause, priority, and missed-epoch treatment
Contract-wrapped deferred compensation	An off-chain legal right that mirrors the mechanism	Employment, tax, accounting, withholding, and enforceability work
Reserve-backed or partially prefunded TLY	A claim supported by disclosed reserves or escrow policy	Reserve governance, coverage reporting, and drawdown rules

The mechanism is legible on paper much earlier than it is defensible in production. The wrapper determines whether alumni hold a weak contingent benefit, a stronger contractual right, or something closer to a prefunded reserve claim.

What Is Actually New Here?

TLY's components are familiar. Deferred compensation, phantom participation, contingent benefits, runoff liabilities, and bonus pools all cover nearby ground. The novelty is architectural: TLY links a historical compensation base to a finite, tapering post-exit cash-flow claim.

- active compensation updates a historical compensation base;
- the exit claim is formulaic rather than negotiated from scratch;
- the post-exit claim is money-denominated, tapering, and finite;
- the claim can be implemented with per-wallet state and pull-based settlement;
- treasury stress can be expressed as an explicit priority rule rather than hidden in token-price volatility.

That combination is the research object. It makes a deferred-compensation promise legible, programmable, and modelable under defined assumptions.

Why Not Simpler Hybrids?

A simpler salary-plus-bonus-plus-deferred-pool design may be better for many organizations. So may RSUs, phantom equity, SARs, standard equity, or token vesting. The case for TLY is strongest when the organization has sufficient operating cash flow, reserve discipline, and governance maturity to support a formulaic continuation claim that does not transfer governance and can be modeled as a runoff burden.

Mechanism	What it does well	Where TLY differs
Salary + bonus + deferred pool	Simple, familiar, easy to document	TLY makes the post-exit runoff formula explicit and contributor-specific
RSUs	Familiar equity-linked compensation for later-stage companies	TLY is cash-flow based rather than share-settlement based
Phantom equity / SAR	Mirrors company value without issuing stock	TLY avoids valuation fights by paying from a treasury reserve asset
Token vesting	Liquid and crypto-native	TLY avoids paying labor through governance-token sell pressure
Standard equity/options	Strong upside and familiar investor logic	TLY gives clearer departure economics but less open-ended upside
Revenue share	Ties payment to organization performance	TLY is based on compensation history, not a direct revenue percentage

If a simpler hybrid gives workers the same clarity with less machinery, use the simpler hybrid. The reason to use TLY is not elegance. It is the specific mix of liquidity, non-dilutive economic continuation, and modelable exit economics.

Comparison With Adjacent Mechanisms

Feature	Standard equity/options	RSUs	Phantom equity / SAR	Deferred bonus pool	Token vesting	TLY
Liquid while active	Usually no	Usually no	Usually no	Usually yes for salary	Usually yes	Yes, for base and active pay
Direct governance dilution	Often yes	Often yes	No	No	Yes	No

Feature	Standard equity/options	RSUs	Phantom equity / SAR	Deferred bonus pool	Token vesting	TLY
Near-term cash obligation	Low	Low to medium	Usually later	Yes	Often indirect	Yes
Retained economics after exit	Sometimes, often illiquid	Plan-specific	Usually yes	Plan-specific	Yes	Yes
Market-price dependence	High	High	High	Low	High	Low if paid in stablecoins
Treasury-solvency dependence	Indirect	Indirect	Medium to high	High	Indirect	High
Fit for pre-revenue startups	Strong	Limited	Limited	Possible if deferred	Common but volatile	Limited unless prefunded, delayed, or threshold-activated

TLY is not categorically better than these mechanisms. It is aimed at a fairly specific problem: giving contributors live, money-denominated upside without leaning on governance-token issuance and without pretending the organization can promise a perpetual tail.

Equity may still be superior for early-stage startups that need pure upside without near-term cash obligations. TLY is better understood as a potential substitute or complement where operating cash flow, reserve discipline, and governance maturity are already present or plausibly reachable.

Where TLY Fits Best

TLY makes the most sense where treasury capacity is already real or at least plausibly reachable. That sounds almost too obvious to write down, but it needs to be written down because mechanism papers have a habit of sneaking past the funding question and acting like good equations fund themselves.

Good candidates include:

- mature DAOs with recurring protocol revenue;
- profitable crypto-native firms;

- protocol-adjacent service organizations;
- cooperatives or mission-driven organizations with reserve discipline;
- treasury-backed ecosystems that want alumni alignment without token sell pressure.

Poor candidates include:

- pre-revenue startups seeking immediate live-payout TLY, unless prefunded, delayed, or threshold-activated;
- organizations with highly volatile treasuries and no stable reserve policy;
- teams unwilling to formalize compensation governance;
- firms that need pure upside asymmetry rather than bounded cash obligations;
- organizations that cannot manage jurisdiction-specific labor, tax, and deferred-compensation requirements.

Funding Models For TLY

Pre-revenue teams are where this gets dangerous fastest. Writing the claim on-chain is trivial compared with funding it later, and teams without a believable treasury path should say that plainly rather than bury it under future-growth assumptions.

Possible funding models include:

1. Operating cash-flow funding: each epoch's claims are paid from ordinary operating cash flow.
2. Margin-linked funding: claim capacity scales with gross margin or operating surplus.
3. Reserve-buffer hybrid: a portion of active payroll, margin, or revenue funds a reserve buffer for future trailing claims.
4. Threshold-activated payout: legacy payouts begin only after revenue, treasury, runway, or reserve-coverage thresholds are met.
5. Waterfall priority model: active pay, operating obligations, reserves, and legacy claims are paid in a disclosed order of priority.

These models can be combined. For example, an early protocol might accrue TLY internally but delay claim activation until a stablecoin reserve threshold is met. That may sound less elegant than a fully live system from day one, but it is often the more honest design. The real question is not whether the claim can be written on-chain. It is whether the organization has a credible funding policy and a disclosed stress rule.

Funding policy determines whether TLY is merely formulaic, credibly reserved, contract-wrapped, or only conditionally claimable. The mechanism does not choose among those variants by itself.

Risks, Limitations, And Failure Modes

Treasury Solvency And The Cash-Flow Paradox

TLY is not for cash-starved organizations with no realistic treasury path. It works best for profitable, treasury-backed, or reserve-capitalized entities. Pre-revenue organizations may need delayed activation, reserve escrows, smaller initial accrual rates, or hybrid token-cash structures. A protocol with unstable revenue but a liquid governance token has not solved the cash problem; it has moved it.

Trailing claims should be described precisely. In the reference on-chain design, they are contingent claims that sit behind active compensation during stress unless an off-chain agreement grants claimants different rights. That is not a small drafting detail. It changes what workers are actually being offered. If treasury funds are insufficient, the organization must define whether missed claims accrue, queue, partially defer, or remain claimable only when the treasury is funded.

Legal, Tax, And Deferred Compensation Risk

TLY may trigger deferred compensation, payroll, securities, labor, tax, employment-benefit, or accounting issues depending on jurisdiction. In some jurisdictions, the legacy claim may need to be implemented through an off-chain employment, contractor, deferred-compensation, or bonus-rights agreement rather than treated as a purely on-chain entitlement. Actual deployment may require wrappers, employer-of-record structures, written off-chain agreements, entity design, and tax withholding processes.

This paper is a mechanism design proposal, not legal, tax, accounting, or investment advice.

Compensation-Ledger Integrity

The natural TLY definition removes the usual terminal-period manipulation problem. The initial trailing claim is not supposed to be based on the contributor's final salary, a final-period bonus, or a number selected at departure. It is the accrual share multiplied by the contributor's eligible lifetime compensation base at the company.

That does not make governance irrelevant. It moves the problem to the place where it belongs: the definition and administration of eligible compensation. If a DAO admin can retroactively edit compensation history, classify a one-off grant as eligible compensation without review, or change the accrual rule after the fact, then the mechanism can still be abused. The right defenses are:

- a clear eligible-compensation definition before accrual begins;
- append-only or auditable compensation-ledger updates;
- review of exceptional compensation and retroactive adjustments;
- role separation for contributor administration, treasury control, and parameter changes;

- reporting that lets contributors and governance see the historical compensation base that will drive future claims.

The current reference contract stores `lastActiveBonus` for simplicity and gas economy. Economically, that value should be read as a cached accrual-rate result derived from historical compensation, not as a discretionary terminal bonus.

Treasury Stress And Insolvency

The Solidity reference includes `pauseLegacyClaims()` so a DAO admin can block legacy withdrawals while preserving active compensation. There is nothing fancy about this. It is just a hard priority rule. It is also not a complete insolvency framework.

Before deployment, organizations should define:

- whether paused claims accrue or simply wait;
- whether claimants can later catch up using aggregate claims;
- whether partial payment is allowed;
- whether legacy claims are legally enforceable debt or contingent protocol benefits;
- what disclosures contributors receive before accepting the structure.

Governance Abuse And Favoritism

TLY depends on credible compensation administration. A pod admin that can update base pay or time departures can influence future trailing claims. If that sounds like a governance headache, that is because it is one.

Mitigations include:

- transparent compensation policies;
- multisig-controlled contributor administration;
- published parameter changes;
- auditable events for base pay updates and exits;
- notice periods and review windows near departure;
- periodic reporting of active payroll, active bonuses, and projected trailing burden.

Parameter Risk

The mechanism is bounded only under coherent parameters. A high accrual rate, weak taper, very long duration, or low-growth payroll can create excessive cash obligations. A 1 percent accrual share sounds small until it compounds through a large historical compensation base under weak turnover assumptions. The simulator should be used before governance adopts or changes parameters.

Implementation Path And Pilot Model

A useful pilot can be small and explicit:

1. Choose the payment asset, usually a stablecoin or treasury reserve asset.
2. Choose epoch length, such as monthly or annual.
3. Choose an initial accrual share, such as 1 percent.
4. Choose a trailing duration, such as 25 annual epochs or 300 monthly epochs.
5. Convert the annual taper into a per-epoch WAD factor.
6. Simulate expected burden under growth, wage, turnover, revenue, and margin assumptions.
7. Define treasury funding policy and stress rules.
8. Define compensation governance, including pod admin authority and review rules near departure.
9. Deploy contracts or begin with an off-chain legal agreement that mirrors the same equations.
10. Publish recurring reports comparing actual burden against simulated burden.

The repository contains two practical tools:

- `sim/app.py`: Streamlit dashboard for scenario analysis.
- `dao/contracts/`: EVM-compatible reference contracts for registry and treasury distribution.

Legal And Regulatory Disclaimer

This document is for mechanism design and technical research only. It is not legal, tax, accounting, employment, securities, benefits, or investment advice. On-chain enforceability does not eliminate off-chain legal obligations. Organizations considering TLY should obtain jurisdiction-specific advice before deployment.

Conclusion

TLY is best understood as a layered compensation architecture. Pure TLY defines the formula. The Stress Layer asks whether the organization can fund it and what happens when it cannot. The Governance Wrapper decides whether the formula is robust enough to survive real people with incentives.

That framing keeps the claim in proportion. TLY can separate contributor upside from governance-asset dilution, offer more liquidity than lottery-style equity, and make departure economics clearer than many option or token packages. None of that makes the claim cheap or easy. It still has to be funded, documented, governed, and legally wrapped in a way that matches the jurisdiction.

If an organization actually has reserve discipline, TLY at least makes the trade legible: contributors get a real trailing claim, and the organization can model the burden before pretending it is manageable.

Appendix A: Full Formulas

Lowercase subscripts refer to contributor-level quantities. Aggregate quantities omit contributor indexing. Unless otherwise stated, all quantities are per epoch.

A.1 Notation

- $B_{i,t}$: contributor i's base pay in epoch t.
- $H_{i,t}$: contributor i's historical compensation base entering epoch t, defined in the reference model as cumulative settled active compensation while active. In the contracts this is stored as `historicalPayPool`.
- $E_{i,t}$: eligible lifetime compensation base used for accrual in epoch t. Under the reference timing convention, $E_{i,t} = H_{i,t} + B_{i,t}$.
- α: active accrual share.
- $A_{i,t}$: active bonus.
- R_i: initial trailing amount after exit.
- s: elapsed epochs since departure, starting at $s = 1$ for the first claimable legacy epoch.
- δ: taper rate.
- $\rho = 1 - \delta$: per-epoch taper factor.
- K: maximum trailing duration.
- q: turnover rate.
- G: gross payroll growth factor.
- L_t: aggregate trailing payout in epoch t.

A.2 Active State

Contributor-level active bonus:

$$A_{i,t} = \alpha E_{i,t} = \alpha(H_{i,t} + B_{i,t}) .$$

Contributor-level historical-pool update:

$$H_{i,t+1} = H_{i,t} + B_{i,t} + A_{i,t} .$$

Aggregate simulator update with turnover:

$$H_{t+1} = (H_t + B_t + A_t)(1 - q) .$$

A.3 Legacy State

Exit compensation-base rule:

$$R_i = \alpha E_{i,\tau_i} .$$

Here E_{i,τ_i} is the contributor's eligible lifetime compensation base at exit under the disclosed timing convention. In the Solidity reference, `lastActiveBonus` caches the same accrual-rate calculation for the last settled active epoch. That cache is an implementation shortcut, not a separate discretionary exit bonus.

This paper uses the EVM timing convention throughout: the first claimable legacy payout occurs one full epoch after departure, so $s = 1$ at the first claimable epoch.

Trailing payout s epochs after departure:

$$Y_{i,s} = R_i \rho^s .$$

Total finite-horizon trailing payout:

$$\sum_{s=1}^{K} R_i \rho^s = R_i \rho \frac{1-\rho^K}{1-\rho} .$$

Infinite-horizon upper bound:

$$\sum_{s=1}^{\infty} R_i \rho^s = R_i \frac{\rho}{1-\rho} .$$

A.4 Aggregate Burden Approximation

New legacy cohort from turnover:

$$N_t = qA_t .$$

Aggregate trailing payout:

$$L_t = \sum_{j=1}^{K} N_{t-j} \rho^{j-1} .$$

Payroll growth:

$$B_{t+1} = GB_t .$$

Active bonus ratio:

$$m_t = \frac{A_t}{B_t} .$$

Historical-pool ratio:

$$h = \frac{H_t}{B_t} .$$

Steady-state relationship:

$$h = \frac{(1-q)(1+\alpha)}{G}(h+1)\,.$$

Let:

$$c = \frac{(1-q)(1+\alpha)}{G}\,.$$

If $c < 1$:

$$h = \frac{c}{1-c}$$

and:

$$m = \frac{\alpha}{1-c}\,.$$

Approximate burden ratio:

$$\frac{L_t}{B_t} \approx qm \sum_{j=1}^{K} \frac{\rho^{j-1}}{G^j}\,.$$

Infinite-horizon upper bound:

$$\frac{L_t}{B_t} \leq \frac{qm}{G-\rho}\,.$$

If the first post-departure claim receives $R_i\rho$, as in the EVM convention:

$$\frac{L_t}{B_t} \leq \frac{qm\rho}{G-\rho}\,.$$

Appendix B: Simulation Assumptions

The Python simulator in `sim/app.py` and `sim/baseline_dollars.py` models the macro version of TLY using cohorts.

Core inputs include:

- `horizon_periods`;
- `initial_workers`;
- `worker_growth`;
- `wage_per_worker`;
- `wage_growth`;
- `accrual_share`;
- `turnover_rate`;
- `legacy_payout_years`;

- `annuity_decay_rate`, which corresponds to the TLY taper rate;
- `revenue_multiplier`;
- `gross_margin`.

The simulator computes:

```
base_payroll = workers * wage
active_bonus = (historical_pay_pool + base_payroll) * accrual_share
legacy_payout = cohort_amount * (1 - taper_rate) ^ cohort_age
historical_pay_pool_next =
    (historical_pay_pool + base_payroll + active_bonus) * (1 - turnover_rate)
```

The simulator is intentionally stylized. It does not model legal enforceability, tax treatment, default litigation, stablecoin depegging, claimant mortality, or endogenous contributor behavior. It is a planning tool for burden dynamics, not a solvency guarantee.

Appendix C: Smart Contract Architecture

The reference EVM implementation lives in `dao/contracts`.

C.1 Contributor Registry

`ContributorRegistry.sol` stores:

- contributor status: none, active, legacy, removed;
- payout address;
- base pay per epoch;
- historical compensation base (`historicalPayPool`);
- cached last active bonus;
- initial legacy amount;
- joined epoch;
- last settled active epoch;
- departure epoch;
- last legacy claim epoch.

`POD_ADMIN_ROLE` can:

- add contributors;
- update payout addresses;
- update base pay;
- remove contributors into legacy status.

Only the configured treasury distributor can settle active epochs and mark legacy claims as paid.

C.2 Treasury Distributor

`TreasuryDistributor.sol` holds the payment token and computes claims.

Active claim:

```
activeBonus =
    (state.historicalPayPool +
state.basePayPerEpoch).mulWadDown(accrualShareWad);
```

Legacy claim:

```
uint256 taperMultiplierWad =
    WadMath.rpowWad(taperFactorPerEpochWad, elapsedEpochs);

claimable += state.initialLegacyAmount.mulWadDown(taperMultiplierWad);
```

The treasury uses OpenZeppelin `AccessControl`, `SafeERC20`, and `ReentrancyGuard`.

C.3 Pull Claims And Gas Safety

The system uses pull claims:

- `claimActiveComp()` for active contributors;
- `claimTrailingYield()` for legacy contributors.

Legacy missed-epoch claims are processed oldest-first and capped by `maxEpochsPerClaim`. This makes transaction cost independent of the total number of alumni.

C.4 Tranche Priority

`pauseLegacyClaims()` blocks legacy withdrawals while preserving active compensation. This supports active-pay priority during treasury stress.

Appendix D: FAQ And Objections

Is TLY just a pension?

No. A pension is typically a broad retirement benefit with employer-specific legal and actuarial treatment. TLY is a contributor-specific trailing cash-flow claim based on lifetime eligible compensation. It tapers and expires.

Is TLY just revenue share?

No. A generic revenue share usually pays a percentage of revenue. TLY pays a defined trailing amount based on contributor compensation history. It may be funded by revenue, reserves, margin, or a hybrid policy, but the claim formula is not a direct revenue percentage.

Why not just give equity?

Equity may be better for early-stage startups that need pure upside and cannot support cash obligations. TLY is better suited to organizations that want liquid, stablecoin-denominated labor upside without transferring governance or ownership.

Why not just use token vesting?

Token vesting is liquid and simple, but it ties compensation to token price and can create sell pressure on the governance asset. TLY can reduce that coupling by paying in a reserve asset.

What if the treasury fails?

TLY depends on treasury solvency. The reference implementation includes active-pay priority and pausable legacy claims, but this does not eliminate funding risk. Organizations must define whether claims are contingent, queued, partially paid, or legally enforceable.

Can someone game the exit bonus?

Not by manipulating a terminal salary number, if TLY is implemented according to the natural definition. The initial trailing amount is the accrual share multiplied by lifetime eligible compensation at the company. Governance risk instead sits in the compensation ledger: what counts as eligible compensation, who can update it, and how exceptional or retroactive compensation is reviewed.

Does this create tax or legal issues?

Potentially. TLY may implicate deferred compensation, payroll, securities, labor, tax, benefits, or accounting rules. Legal design is jurisdiction- specific.

Why not cap historical compensation?

The reference mechanism intentionally avoids artificial caps on the historical compensation base because long-tenure contributors are supposed to earn larger trailing claims. Aggregate stability comes from low active accrual, payroll growth, finite duration, and tapering legacy obligations. Governance may still restrict manipulation of the eligible-compensation ledger.